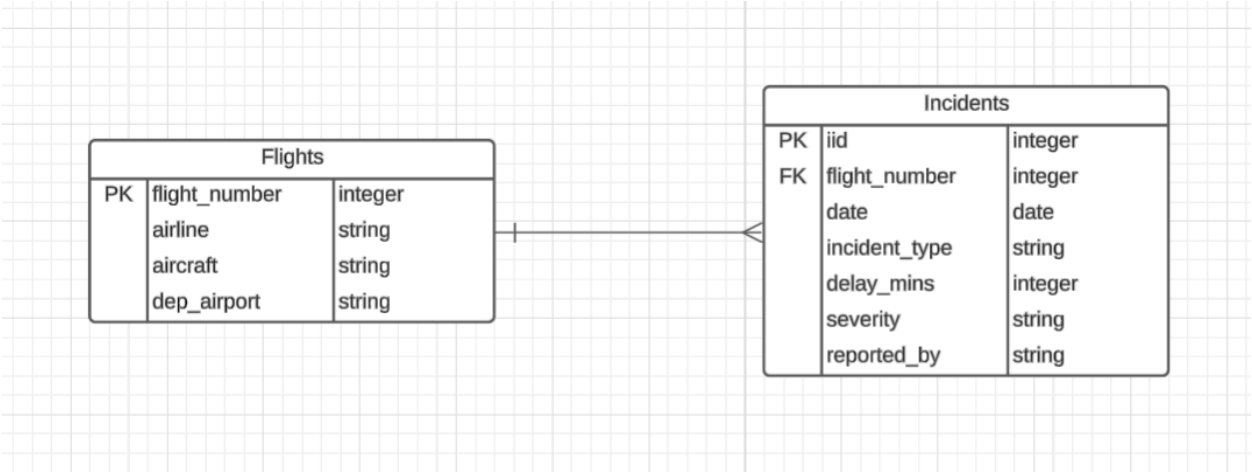
Diego Cicotoste & Sarah Taylor
CS3200 - Summer I, 2025
June 16, 2025
Final Project - Part B

1. For the relation represented by all of the columns in the CSV file, define all functional dependencies and list them.
    - Inside of the CSV it contained the following dependencies:
        - airline, aircraft, and dep_airport were dependent on flight_number
        - flight number, date, incident_type, severity, delay_mins, num_injuries, and reported_by were dependent on iid.

2. Using the functional dependencies and the rules of normalization, decompose the relational from the CSV into several relations that all satisfy 3NF; give the relations reasonable names.
    - The functional dependencies got separated into 2 tables
        - Flights -> (PK) flight_number, airline, aircraft, dep_airport
        - Incidents -> (PK) iid, (FK) flight number, date, incident_type, severity, delay_mins, num_injuries, reported_by
3. For the relations resulting from the normalization, create an ERD in the IE (Crow's Feet) notation. Add all attributes, attribute name, primary and foreign keys, data types, and entity descriptions.



- **1NF:** Each row has only one value - nothing repeated
- **2NF**: The key is just iid, so there is no partial dependency
- **3NF:** No other row depends on anything except the key
- and all the attributes in each relation depend only on the primary key and not on any non-key attribute.

Because all of these statements are true this is a 3NF